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Development of AI-Based Digital Book Publishing Application For Ethiopian Authors and Readers, by 2026

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A Project Report Submitted in Partial Fulfilment of the Requirements for the Degree of Bachelor of Science in Computer Science

Debre Berhan, Ethiopia
December 2024

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CHAPTER ONE

1. INTRODUCTION

1.1 Background of The Project

The advancement of information and communication technology has significantly transformed the publishing industry worldwide. Digital publishing platforms have enabled authors to publish their works online and readers to access content conveniently through various digital devices. These platforms have reduced the cost of publishing, increased content accessibility, and expanded opportunity for authors to reach wider audiences.

Artificial Intelligence (AI) has further enhanced digital publishing by providing intelligent writing assistance and personalized recommendation system. AI-powered technologies can assist readers discover books that match their interests and preference. As a result,digital publishing ecosystem have become more efficient, interactive, and user-centered.

Despite these technological advancements, Ethiopia is still faces significant challenges in the digital publishing sector. Most existing digital publishing platforms are designed for international audience and provide limited support for Ethiopian languages, particularly Amharic. Ethiopian \authors often struggled to publish and distribute their books digitally, while readers face difficulties accessing local books through convenient online platforms. Furthermore, there is a lack of integrated system that support Ethiopian book publishing, purchasing, recommendation services, and offline reading capabilities.

Many Ethiopian readers rely on physical books, which may be expensive, difficult to access, or unavailable in certain location. Existing digital reading platforms also require continuous internet access, making them less practical in areas with limited connectivity. Additionally, there is no comprehensive platform specifically designed to support Ethiopian authors, Ethiopian readers,and Amharic-languages content using modern AI technologies.

To address these challenges, this project proposes the design and implementation of GETSE, an AI-Driven digital publishing ecosystem for Ethiopian authors and readers with intelligent personalized recommendation. The system will consist of a web platform and mobile application. The web platform will allow authors to publish books, receive AI-assisted writing support, and receive personalized recommendations for books. The mobile application enable users to download purchased books and read the offline, reducing dependency on internet connectivity while promoting digital reading culture in Ethiopia.

1.2 Problem Statement and Justification

The Ethiopian publishing industry faces several challenges related to digital transformation and accessibility.
The major problems include:

- Lack of a dedicated Ethiopian digital publishing platform
- Limited support for Amharic and other ethiopian languages in existing publishong system
- Difficulty for ethiopian authors and distribute books digitally.

- Limited visibility of Ethiopian books in international digital marketplaces.
- Lack of intelligent recommendation systems tailored to ethiopian readers.
- Dependence on physical books that may be expensive or difficult to obtain.
- Lack of integrated platforms that combine publishing purcahsing recommendation and offline reading services.

These challenges limit opportunities for authors, reduce accessibility for readers, and slow the digital transformation of Ethiopia's publishing industry. Therefore, the development of GETSE is justified as it provides and integrated platform that promotes Ethiopian literature, supports local language, enhances content discovery , and improves access to books through digital technologies.

1.3 Project Objectives

1.3.1 General Objective

To design and implement GETSE, an AI-Driven digital publishing ecosystem that supports Ethiopian authors and readers through personalized recommendation, digital publishing, online purchasing, and offline reading services.

1.3.2 Specific Objectives

- To analyze the requirements of an Ethiopian digital publishing ecosystem.
- To design and develop a web platform for ethiopian authors and readers.
- To implement support for amharic and english language.
- To implement a personalized recommendaton system for readers.
- To develop a digital book publishing and management sysetem.
- To implement online book search and purcchasing functionalites.
- To develop secure user registration and authentication mechanisms.
- To design and develop a mobile application for offline reading of purchased books.
- To test and evaluate the usability, performance, and effectiveness of the developed system.

1.4 Project Scope and Limitations

1.4.1 Scope

The project focuses on developing an integrated digital publishing ecosystem specifically designed for ethiopian authors and readers.
The system will provide:

- User registration and login.
- Author profile management.
- Reader profile mangement.
- Book uploding and publishing services.
- Amraric language support.
- Personalized book recommendation services.
- Book searching and filtering functionlities.

- Online book purchasing.
- Digital library management.
- Mobile application for offine reading.
- Administrative management functionalities.

Target users inclide:

- Ethiopian authors.
- Ethiopian readers.
- System adminiatrators.

This project under the category of web application and mobile application development integrated with artificial intelligence technologies.

1.4.2 Limitations

The project may have the following limitations:

- Recommendation quality depends on available user interaction data.
- The system may initially support only selected Ethiopian languages primarily Amharic.
- Copyright verification may not be fully automated.
- The system will focus on digital books and will not include physical book delivery services.
- Offline reading functionality will not be available for purchased books.

1.5 Significance of The Project

For Ethiopian authors

- Provide a dedicated platform for publishing digital books
- Increase visibility of Ethiopian literary works.
- Create opportunites for generating income from digital publishing.

For ethiopian readers

- Provides easy access to Ethiopian books.
- Supports personilized reading exeperiences.
- Enable offline reading through moblie devices.
- Promotes reading culture through digital accessibility.

For the Ethiopian publishing industry

- Supports digital transformation of publishing services.

- Encourages the adoption of modern publishing technologies.
- Expands the market research of local authors and publishers.

For society

- Promotes Ethiopian literature and culture
- Encourages knoledge sharing and lifelong learning.
- Improves access to educational and literary resources across Ethiopian.

CHAPTER TWO

2. REVIEW OF RELATED WORK

2.1 Introduction

This chapter reviews existing digital publishing systems, recommendation systems, and digital reading platforms related to the proposed project. The review identifies strengths and limitations of existing solutions and highlights the research gap that motivates the development of GETSE.

2.2 Existing Digital Publishing Platforms

2.2.1 Amazon Kindle Direct Publishing (KDP)

Amazon kindle direct publishing enables authors to self-publish and distribute digital books globally.

Strength

- Large international audience.
- Easy publishing process.
- Secure digital distribution.
- Revenue generation opportunities.

Limitations

- Limited support for Ethiopian publishing needs.
- No specialized support for Amharic content.
- Focuses primarily on international market.

2.2.2 Wattpad

Wattpad is a storytelling platform that allows authors to publish and share content online.

Strength

- Interactive community.
- Easy content publishing.

- Reader engagement features.
- Supports emerging writers.

Limitations

- Limited recommendation accuracy.
- Lack of integrated AI writing support.
- No specialized support for Ethiopian authors and readers.
- Limited monetization.

2.2.3 Medium

Medium is an online publishing platform for articles and blogs.

Strength

- User-friendly interface.
- Large readership.
- Content categorization capabilities.

Limitations

- Not designed for book publishing.
- Limited personalization features.
- No dedicated Ethiopian content ecosystem.

2.3 AI-Assisted Publishing Support Systems

AI-powered publishing support system assist authors during the publishing process by providing book reviews, publication recommendations, content analysis, and cover-page design suggestions.

Strengths

- Provide feedback and recommendations for book improvement.
- Support content review and quality assessment.
- Suggests suitable cover page concepts based on the book's theme and description.
- Improve publishing efficiency and user experience.

Limitations

- Recommendations may require authors evaluation and modification.

- Quality of suggestions depends on the information provided by the author.
- Limited support for Ethiopian publishing contexts in existing systems.
- Insufficient optimization for Amharic-language content.
- Cover design suggestions may require further customization by the author.

2.4 Recommendation Systems

Recommendation systems help users discover relevant content based on preferences and behavior.

Content-based filtering

Strength

- Personalized suggestions.
- Easy implementation.
- Uses content characteristics effectively.

Limitations

- Limited content discovery.
- Difficulty recommending new categories.

Collaborative filtering

Strength

- Discovers new content.
- Provides accurate recommendations when sufficient data exists.

Limitations

- Cold-start problem.
- Requires large amounts of user data.

2.5 Mobile Reading Applications

Digital applications enable users to access books through smart-phones and tablets.

Strength

- Convenient reading experience.
- Offline accessibility.
- Portable digital library.

Limitations

- Often separated from publishing platforms.
- Limited integration with recommendation system.
- Lack of Ethiopian-language-focused content ecosystems.

2.6 Comparative Analysis

Table 1: Comparative analysis

Feature	KDP	Wattpad	Medium	Proposed GETSE
Digital Publishing	Yes	Yes	Limited	Yes
AI Content Generation	No	No	No	Yes
Personalized Recommendation	Partial	Partial	Limited	Yes
Ethiopian Language Support	No	No	No	Yes
Amharic Content Support	No	No	No	Yes
Online Book Purchase	Yes	Limited	No	Yes
Offline Reading	Partial	Partial	Partial	Yes
Mobile Application	Yes	Yes	Yes	Yes
Ethiopian Publishing Focus	No	No	No	Yes

2.7 Research Gap

The reviewed system provide publishing and reading services but do not adequately address the needs of Ethiopian authors and readers. Most platforms lack support for Amharic-language content, Ethiopian book publishing workflows, localized recommendation systems, and integrated AI-assisted writing support.

Furthermore, there is no unified platform that combines Ethiopian digital publishing, intelligent content generation, personalized recommendations, online book purchasing, and offline reading capabilities. This gap creates the need for the proposed GETSE platform.

2.8 Chapter Summary

This chapter reviewed existing digital publishing platforms, recommendation systems, AI-assisted writing technologies, and mobile reading applications. This analysis identified limitations in supporting Ethiopian authors, readers, and local-language content, justifying the development of the proposed GETSE platform.

CHAPTER THREE

3. METHDOLOGY

3.1 Introduction

This chapter presents the methodologies, tools, technologies, techniques and approaches used for the design and implementation of GETSE. It describes the development methodology, data collection methods, system architecture, programming tools, AI technologies and testing approaches selected for the project.

3.2 Development Methodology

The project will adopt Agile Software Development Methodology.
Justification
Agile was selected because:

- It supports itrative development.
- It accommodates changing requirements.
- It promotes continuous feedback.
- It allows frequent testing and improvement.
- It supports incremental delivery of features.

The project will be developed through multiple iterations including planning, design implementation, testing, and evaluation.

3.3 Data Collection Methods

3.3.1 Litrature Review

Online resources related to digital publishing, Artificial Intelligence, recommendation systems and mobile applications will be reviewed.
Justification

Provides technical knowledge necessary for system development.

3.3.2 Existingsystem Analysis

Existing publishing platforms and reading applications will be analyzed.

Justification

Helps identify strengths, weaknesses, and missing features relevant to Ethiopian users.

3.4 System Development Approach

The project follows a full-stack development approach involving:

- Frontend development
- Backend development
- Database management
- AI integration
- Mobile application development

The development stages include:

1. Requirement Analysis
2. System modeling
3. System design
4. Database design
5. System impliementation
6. System testing
7. System evaluation

3.5 System Architecture

The proposed system will utilize Three-Tier Architecture

Presentation Layer

Provides interfaces for authors, readers, and administrators through web and mobile applications.

Application Layer

Handles business logic, recommendation services, authentication, and purchasing operations.

Data Layer

Stores book user profiles, trnsactions, recommendations, and system records.

Justification

Three-tire architecture improves:

- Scalability
- Security
- Maintainability
- System flexibility
- Ease of future enhancements

3.6 Tools and technologies

React.js

Used for web application development.

Justification

- Reusable components.
- High performance.
- Responsive user interfaces.

HTML, CSS, JavaScript

Used for designing and implementing user interfaces.

3.6.2 Backend Technologies

Node.js and Express.js

Used for server side development and API creation.

Justification

- High performance.
- Scalability.
- Fast development
- Excellent API support.
- JavaScript integration.

3.6.3 Database Technology

postgreSQL

Used for data sotarge and management.

Justification

- Reliability
- High performance
- Strong security
- Support for complex queries.

3.6.4 Mobile Application Technology

Flutter

Used for developing the offline reading application.

Justification

- Cross-platform support
- High performance
- Faster development process

3.6.5 Aritificial Intelligence Technologies

Natural language processing (NPL)

Used for:

- Text analysis
- Language processing

Machine Learning

Used for:

- Personalized recommendations.
- User preference analysis
- Reading behavior prediction

Justification

These technologies provide intelligent functionality that improves user experience for both authors and readers.

3.7 System Design Techniques

The following techniques will be used:

Use case Diagram

Models interactions between users and the system.

Activity Diagram

Represents workflow processes.

Sequence Diagram

Illustrates communication between system components.

Entity relationship Diagram(ERD)

Designs the database structure.

Justification

These techniques improve understanding, planning, and implementation of the system.

3.8 Testing Methodology

The following testing methods will be used :

Unit testing

Tests individual modules independently.

Integration testing

Test communication between modules.

System Testing

Tests the complete integrated modules.

User acceptance testing

Ensures that the system satisfies user requirements.

Justification

Testing ensures system quality, reliability, security, and usability.

3.9 Chapter summary

This chapter presented the methodologies, approaches, tools and technologies selected for developing GETSE . Agile methodology, Three-Tier Architecture, React.js, Django, PostgresSQL, Flutter, Machine learning and Natural Language Processing technologies were selected because they provide flexibility, scalability , maintainability and intelligent capabilities required for building an Ethiopian AI-driven digital publishing Ecosystem.

References

[1] K. Pohl, Requirements Engineering: Fundamentals, Principles, and Techniques. Springer, 2010.

[2] R. S. Pressman, Software Engineering: A Practitioner’s Approach.

[3] IEEE Standards Association. https://standards.ieee.org

[4] Amazon Kindle Publishing Guidelines.

[5] Wattpad Platform Documentation.